



LWG Assessment of DOE's Energy Portfolio

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*Basic Energy Sciences Advisory Committee
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Motivation

"We have not done as good a job as we should in coordinating the activities of the ESE offices. We have not done as good a job as we should in performing the crosscutting analysis we need to justify our budgets to the Congress."

David Garman

Under Secretary for Energy, Science and Environment

Senate Confirmation Hearing

April 6, 2005

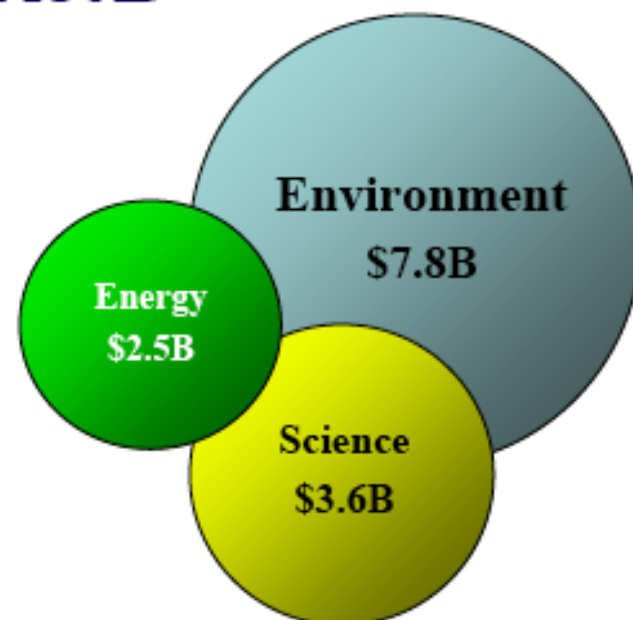


Program Scope

Applied Energy R&D

FY 2005 Appropriation

Applied Energy R&D Portfolio



ESE Portfolio = \$14B

Program Offices

- Energy Efficiency and Renewable Energy (EE)
- Fossil Energy (FE)
- Nuclear Energy, Science, and Technology (NE)
- Electricity Delivery and Energy Reliability (OE)

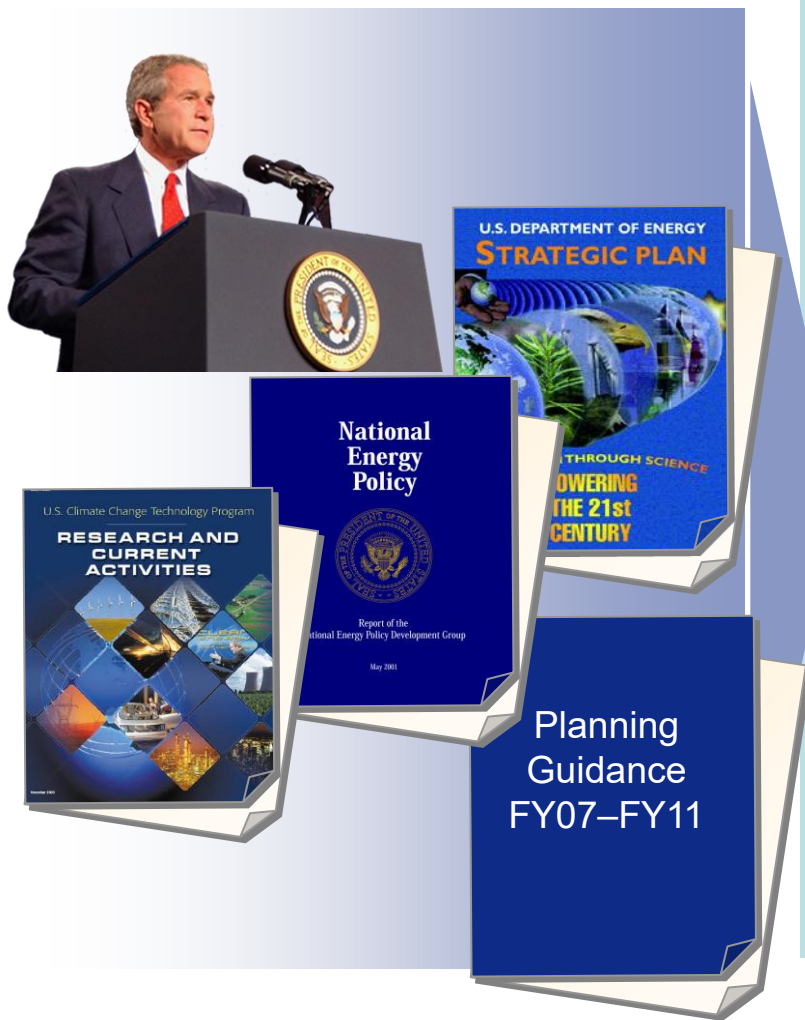
Applied Energy Programs Units

- High Temperature Superconductivity
- Transmission Reliability R&D
- Electric Distribution Transformation R&D
- Energy Storage R&D
- GridWise
- GridWorks
- Infrastructure Security Technology R&D
- Biomass
- Buildings Technologies
- Distributed Energy
- Vehicles Technologies
- Geothermal Technologies
- Hydrogen, Fuel Cells, and Infrastructure Technologies
- Industrial Technologies
- Solar Energy Technologies
- Wind Technologies
- Coal and Other Power Systems
- Nuclear Power 2010
- Generation IV Nuclear Energy
- Nuclear Hydrogen Initiative
- Advanced Fuel Cycle Initiative

Charge to Laboratory Working Group (LWG)

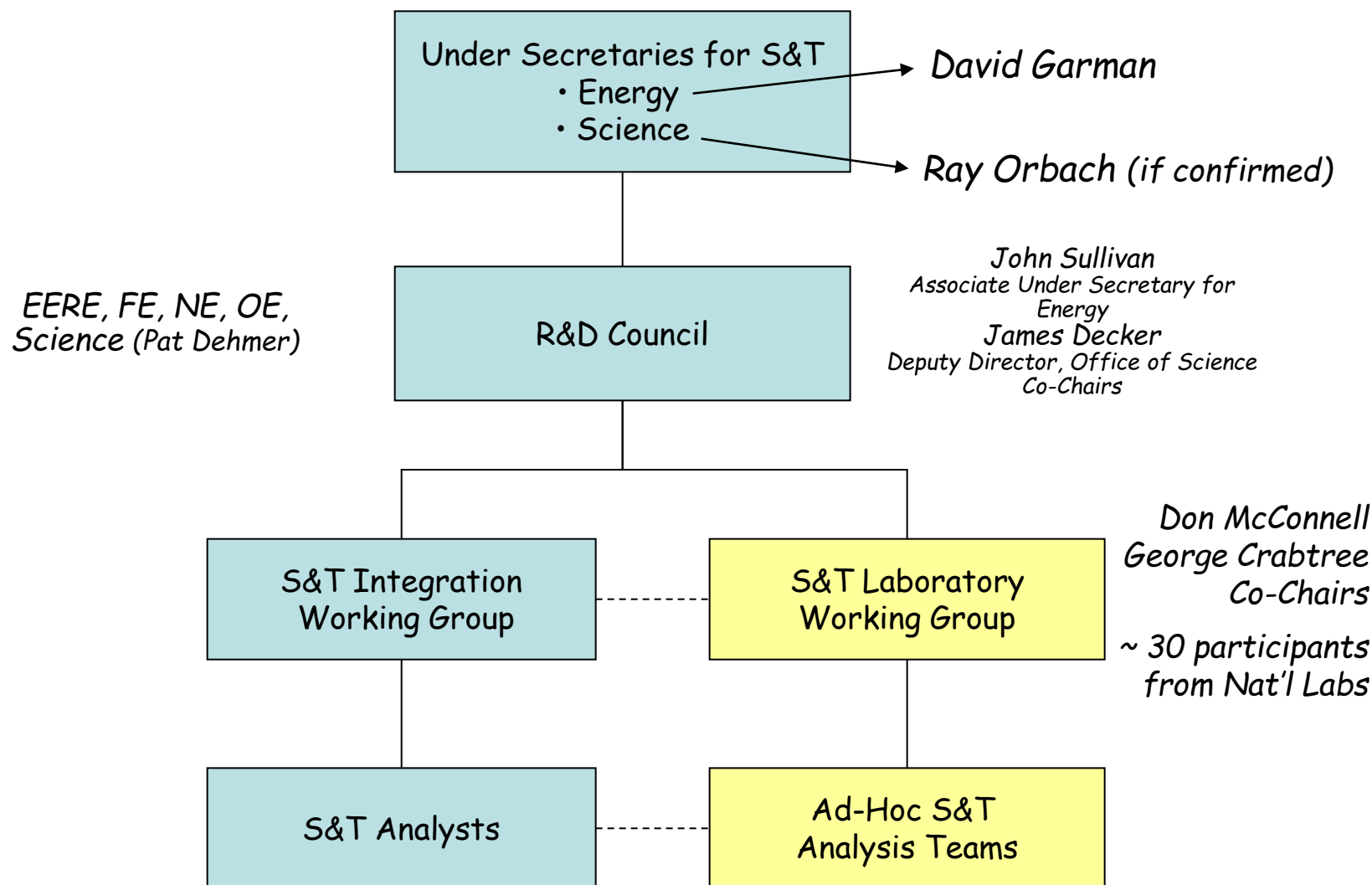
- Deliver improved analysis and decision support material to senior management for the FY08 budget process through a focused effort
 - Emphasis on simplicity, timeliness, clarity, and relevance
 - Focus on impacts and risks
 - Draw on best available knowledge and capability from DOE and the national laboratories
- Develop a sustainable long-term portfolio analysis capability and process
 - Set the standard for credibility
 - Incorporate technical, market, and other risks
 - Enable timely response to changing circumstances

The Context: Advancing Four, Broad National Energy Policy Goals



1. **Diversify our energy mix and reduce dependence on foreign petroleum**, thereby reducing vulnerability to disruption and increasing the flexibility of the market to meet U.S. needs
2. **Reduce greenhouse gas emissions and other environmental impacts** (water use, land use, criteria pollutants) from our energy production and use
3. **Create a more flexible, more reliable and higher capacity U.S. energy infrastructure**, thereby improving energy services throughout the economy, enabling use of diverse sources, and improving robustness against disruption
4. **Improve the energy productivity** (or energy efficiency) of the U.S. economy

LWG Organization



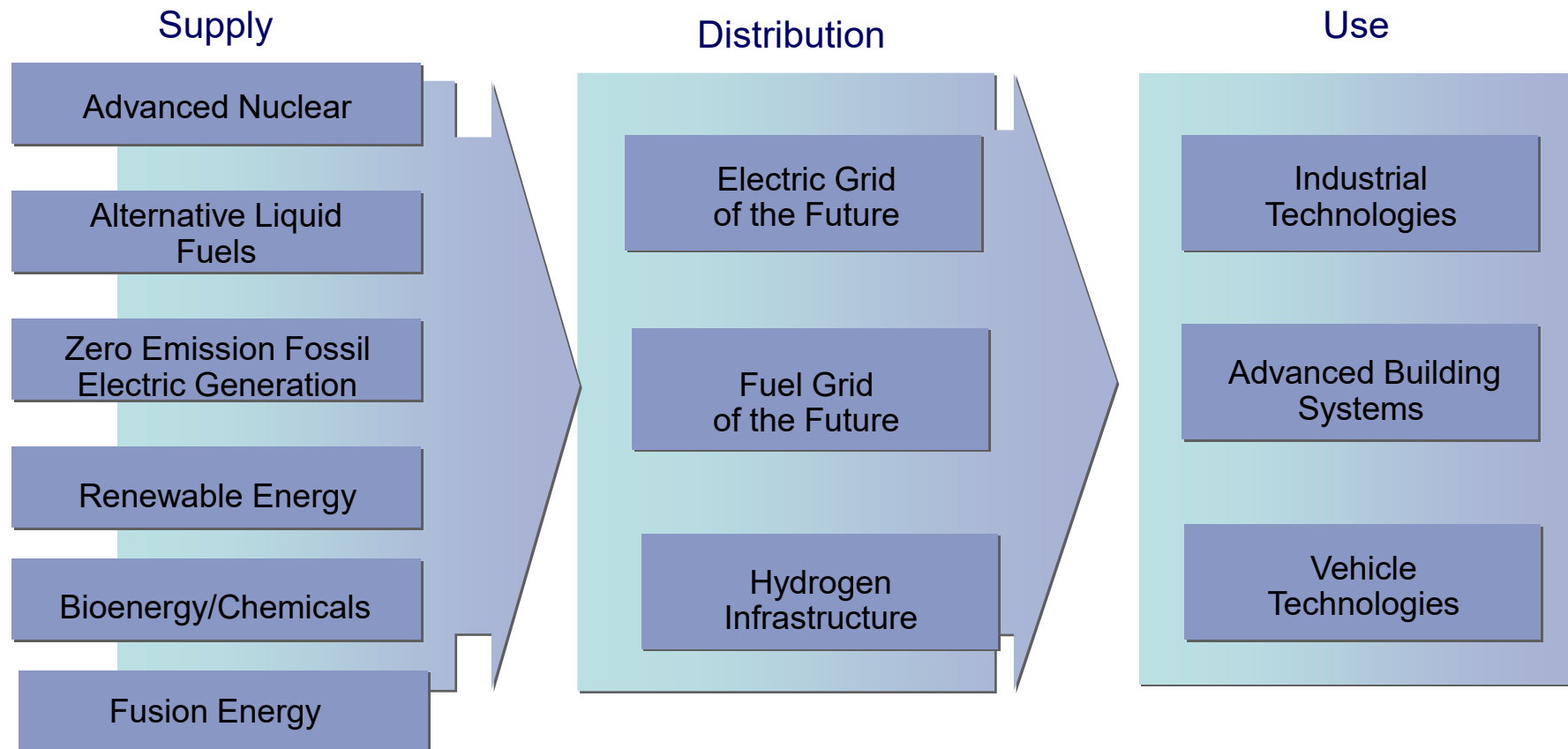
Multi-year Process

FY05 (for FY07 programs)	applied energy programs, qualitative impact
FY06: (for FY08 programs)	+ quantitative impact, relation to science, risk
FY07 (for FY09 cycle)	+ model analysis
FY08 (for FY10 cycle)	

Analysis Tasks

- ✓ Task 1: Energy R&D Innovation Strand
definition, assessment & characterization
- ✓ Task 2: Innovation Strand impact analysis
- Task 3: Integrated portfolio assessment
- Task 4: Recommendations for an enduring S&T assessment
process

Innovation Strands



Future Electricity Systems

Future Liquid Fuels & Transportation

Future Hydrogen & Gas Systems

Cross-cutting / Enabling Science and Technology Opportunities & Challenges

General Observations on the Portfolio
FY05 (for FY07 cycle)

Earmarks and Budget Swings

- Five of the programs have experienced continual upswings during President Bush's first term (see green-shaded).
 - The number drops to 3 when earmarks are included.
- Almost half of the portfolio's programs (10) have experienced consecutive budget up-swing/down-swing years, or vice-versa (see yellow-shaded).
 - The number drops to 9 when earmarks are included.
- Three programs experienced continual downswings during this period (see red-shaded).
 - The number climbs to 6 when earmarks are included.

Programs	Not Including Earmarks		Including Earmarks	
	FY03 to FY04	FY04 to FY05	FY03 to FY04	FY03 to FY05
Transmission Reliability R&D	\$ 3.7	\$ 4.2	\$ 1.5	\$ 0.8
HFCIT	\$ 52.2	\$ 24.8	\$ 10.2	\$ 29.4
Generation IV Nuclear Energy	\$ 12.5	\$ 16.0	\$ (0.5)	\$ 31.0
Nuclear Hydrogen Initiative	\$ 5.0	\$ 2.0	\$ 3.0	\$ 2.0
Advanced Fuel Cycle Initiative	\$ 11.6	\$ -	\$ 51.9	\$ (8.5)
High Temperature Superconductivity	\$ (10.7)	\$ 17.4	\$ (11.7)	\$ 3.9
Electric Distr. Transf. R&D	\$ 6.9	\$ (8.7)	\$ (3.1)	\$ (0.5)
Energy Storage R&D	\$ 1.1	\$ (4.8)	\$ (2.5)	\$ 0.2
Buildings Technologies	\$ (0.5)	\$ 7.7	\$ (0.8)	\$ 7.9
Geothermal Technologies	\$ (3.8)	\$ 0.6	\$ (5.7)	\$ (1.0)
Solar Energy Technologies	\$ (14.5)	\$ 4.3	\$ (15.6)	\$ (4.3)
Wind Technologies	\$ (1.8)	\$ 1.0	\$ (3.3)	\$ (2.1)
Coal and Other Power Systems	\$ 28.7	\$ (87.9)	\$ 28.7	\$ (87.9)
Nuclear Power 2010	\$ (9.9)	\$ 30.0	\$ (8.9)	\$ 30.0
Distributed Energy	\$ (0.4)	\$ 0.7	\$ (0.4)	\$ 0.7
Biomass	\$ (17.4)	\$ (3.5)	\$ (31.7)	\$ 2.4
Vehicles Technologies	\$ (1.8)	\$ (7.0)	\$ (1.8)	\$ (7.0)
Industrial Technologies	\$ (6.4)	\$ (15.6)	\$ (6.4)	\$ (15.6)
GridWise*	\$ -	\$ 6.4	\$ -	\$ 3.4
GridWorks*	\$ -	\$ 5.5	\$ -	\$ 3.5
Infrastructure Security Technology R&D*	\$ -	\$ -	\$ -	\$ -
*Program created in after FY04, not enough data to determine trend.				

***There are several crosscutting technical challenges
that warrant focused attention***

- Energy storage at every scale is a critical issue in multiple technology strands
 - Applies to electric grid, buildings, vehicles, renewables
 - Need is for both high power density and low weight
- Electrochemical conversion (at high and low temperature) is a key issue
 - Applies to hydrogen, fuel cells, energy storage
- New materials for extreme environments are required in multiple technology areas
 - Nuclear power, fusion, hydrogen production
- Real-time adaptive control of large scale or complex systems is required at multiple scales
 - Engines, buildings, electric grid

Several areas of science have particularly high enabling potential

- Nanostructured materials will have transforming impact in the near, mid and long term
 - Energy storage and conversion, solar power, hydrogen storage
 - Engineered materials (e.g. active building components)
 - Materials for extreme environments (especially VHT nuclear)
- Catalysis advances will enable
 - Energy conversion, zero emission hydrocarbons, biomass
- Advances in systems biology can “change the game” for biofuels and bioproducts
 - Engineered feed stocks, bioprocessing technologies
- Advances in high temperature superconductivity are important for both the grid and for fusion
- High end computational modeling and simulation has very high potential for enabling technological advance in many areas
 - Engines, fuel cells, process technologies, efficient power plants, etc.

Basic Science Vision

Incremental advances in the state of the art of existing energy technologies will not meet the nation's future energy and environmental security challenges.

Revolutionary innovations are needed, both in the energy technologies themselves and in our understanding of the fundamental science that enables their operation.

Vibrant fundamental science programs generate revolutionary innovations in two ways: (i) by discovery-driven advances in the frontier of knowledge, enabling new paradigms and unexpected opportunities for disruptive energy technologies, and (ii) by use-inspired research targeting specific areas where incomplete understanding blocks technological progress. DOE should maintain strong programs in both areas that sustain US leadership in science.

Basic-applied interactions are a fertile source of innovation. DOE should develop new ways to stimulate translational research and creative connections across the basic-applied interface.

Basic Science Frontiers

High Performance Materials

Science at the nanoscale, especially low-dimensional systems

Dynamics of physical, chemical and biological phenomena

Emergent behavior in complex systems, from high T_c superconductors to pattern formation in chemical solutions to self-assembly and self-repair

Catalysis and control of chemical transformation

Molecular to systems level understanding of living systems

Biomimetics and photobiological energy conversion

Molecular scale understanding of interfacial science, separations, and permeability in physical systems and membranes

New Tools for:

In situ molecular characterization

Theory/Computation/Numerical Applications

Biomolecule production and characterization

Basic Science Frontiers

High Performance Materials

Research Directions:

- Stability in extreme environments: temperature, corrosion, radiation
- Greater functionality: fast, small, strong, smart, efficient, multifunctional

Scientific Challenges:

- Understand structure-function relationship at all scales
- Simulate/model behavior from first principles
- Create properties through nanoscale design

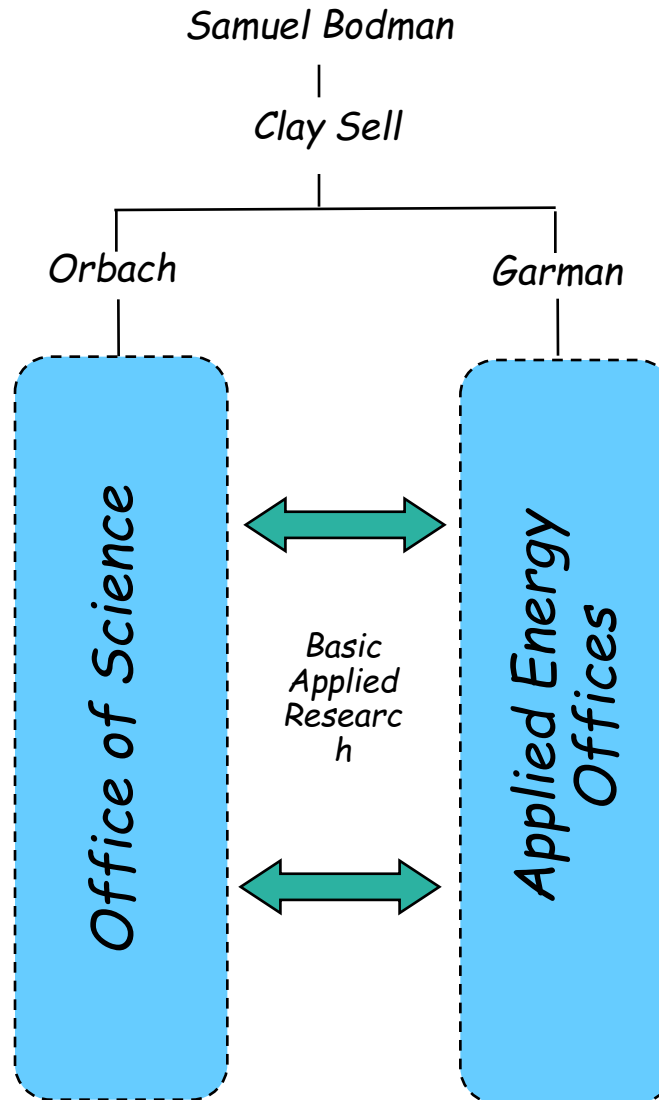
Potential Impact:

Next generation materials for nuclear reactors, high temperature thermochemistry, superconductivity, catalysis, biomimetics, energy conversion among photons, electrons, chemical compounds and heat

Timescale:

Continuous. Advances are interdependent- discovery in one class of materials triggers breakthroughs in another

Basic-Applied Research



Basic-Applied Research

What are the goals?

Translation of applications from basic to applied

50% efficient quantum dot solar cell

Cost competitive superconducting wire

Develop disruptive approach to grand energy challenges

Make an electronic switch $\Rightarrow$ information revolution

Store 24 GWh of electrical energy for 24 hours

Personal transportation at $1/10^{\text{th}}$ cost of cars

What are the attributes?

Integrated basic-applied PI teams

Integrated basic-applied management teams

Tap the best scientists/engineers: innovative thinkers, receptive to new ideas and people

Objectives are innovation driven, not time-scale driven

Stable program: 10+ year life

International network of workshops and visitors to create community and stimulate fresh perspective

Periodic review by top scientists/engineers outside DOE

Examine other innovation machines for organizational inspiration: DARPA, Bell Labs, Google, Microsoft, Apple, Xerox Parc